

**DYNAMICAL SYSTEMS:
CLASS SEPTEMBER 15, 2026
6. 2D NONLINEAR SYSTEMS AND FIXED POINTS**

ROB TOMPKINS

Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- `teamXX-YY.EXT`

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

Team Activity

(1.) (6.4.2) Consider the system

$$\dot{x} = x(3 - 2x - y), \quad \dot{y} = y(2 - x - y), \quad x, y \geq 0.$$

- (a) Find the fixed points.
- (b) Draw the nullclines on the xy -plane.
- (c) Add a representative vector in each region of phase space.
- (d) Compute the Jacobian matrix.
- (e) Classify the fixed points. Compare your classification to the behavior of the vector field near each fixed point in your diagram from part (c).
- (f) Sketch a plausible phase portrait for the system. Compare your sketch to a numerically computed phase portrait.

Solution. Write

$$f(x, y) = x(3 - 2x - y), \quad g(x, y) = y(2 - x - y).$$

- (a) The fixed points are the intersections of

$$x = 0 \quad \text{or} \quad 3 - 2x - y = 0$$

with

$$y = 0 \quad \text{or} \quad 2 - x - y = 0.$$

Thus

$$\boxed{(0, 0), \quad (0, 2), \quad \left(\frac{3}{2}, 0\right), \quad (1, 1)}.$$

For the interior point, subtracting $2x + y = 3$ and $x + y = 2$ gives $x = 1$, hence $y = 1$.

(b) The x -nullclines are

$$x = 0, \quad y = 3 - 2x,$$

and the y -nullclines are

$$y = 0, \quad y = 2 - x.$$

In the first quadrant the oblique x -nullcline runs from $(0, 3)$ to $(3/2, 0)$, while the oblique y -nullcline runs from $(0, 2)$ to $(2, 0)$. They cross at $(1, 1)$.

(c) Away from the axes, the signs are determined by

condition	sign
$y < 3 - 2x$	$\dot{x} > 0$
$y > 3 - 2x$	$\dot{x} < 0$
$y < 2 - x$	$\dot{y} > 0$
$y > 2 - x$	$\dot{y} < 0$

Therefore the four possible directions are $\nearrow$ for $(+, +)$, $\searrow$ for $(+, -)$, $\nwarrow$ for $(-, +)$, and $\swarrow$ for $(-, -)$. For example, representatives are

(x, y)	$(\text{sgn } \dot{x}, \text{sgn } \dot{y})$	direction
$(1/2, 1/2)$	$(+, +)$	$\nearrow$
$(1/2, 7/4)$	$(+, -)$	$\searrow$
$(5/4, 3/5)$	$(-, +)$	$\nwarrow$
$(1/2, 5/2)$	$(-, -)$	$\swarrow$

(d) Differentiating f and g gives

$$J(x, y) = \begin{bmatrix} 3 - 4x - y & -x \\ -y & 2 - x - 2y \end{bmatrix}.$$

(e) Evaluation at the fixed points gives

(x, y)	$J(x, y)$	eigenvalues	classification
$(0, 0)$	$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$	$3, 2$	unstable node
$(0, 2)$	$\begin{bmatrix} 1 & 0 \\ -2 & -2 \end{bmatrix}$	$1, -2$	saddle
$(3/2, 0)$	$\begin{bmatrix} -3 & -3/2 \\ 0 & 1/2 \end{bmatrix}$	$-3, 1/2$	saddle
$(1, 1)$	$\begin{bmatrix} -2 & -1 \\ -1 & -1 \end{bmatrix}$	$\frac{-3 \pm \sqrt{5}}{2}$	stable node

These classifications agree with the local arrow directions from part (c): arrows leave $(0, 0)$, enter and leave each saddle along different directions, and point toward $(1, 1)$ from all nearby sectors.

(f) The coordinate axes are invariant. On the positive x -axis, solutions approach $(3/2, 0)$; on the positive y -axis, solutions approach $(0, 2)$. Interior trajectories cannot cross either axis and are directed toward the stable coexistence equilibrium $(1, 1)$. A numerical phase portrait should therefore show the two boundary saddles, their invariant-axis trajectories, and interior trajectories curving into $(1, 1)$.

□

- (2.) **Analyzing a 2D System:** Consider a 2D red fox–coyote system (let “1” denote red foxes and “2” denote coyotes):

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1}{K_1}\right) - \alpha_1 N_1 N_2, \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2}{K_2}\right) - \alpha_2 N_1 N_2.\end{aligned}$$

- (a) Using $N_1 = A_1 x$, $N_2 = A_2 y$, and $t = T\tau$, show that the system can be nondimensionalized to give

$$\begin{aligned}\frac{dx}{d\tau} &= x(1 - x) - \beta_1 xy, \\ \frac{dy}{d\tau} &= \rho y(1 - y) - \beta_2 xy,\end{aligned}$$

where

$$\rho = \frac{r_2}{r_1}, \quad \beta_1 = \frac{\alpha_1 K_2}{r_1}, \quad \beta_2 = \frac{\alpha_2 K_1}{r_1}.$$

- i. Show that the nullclines of this nondimensional system are

$$\begin{aligned}x \text{ nullclines: } & x = 0, & y &= \frac{1}{\beta_1}(1 - x), \\ y \text{ nullclines: } & y = 0, & y &= 1 - \frac{\beta_2}{\rho}x.\end{aligned}$$

- ii. There are three equilibria whose locations are unaffected by the parameters. What are they? Interpret these equilibria in terms of foxes and coyotes.
iii. Find conditions to ensure that there is exactly one crossing of the nullclines in the first quadrant (not on the axes). How would you interpret these conditions in terms of foxes and coyotes?
- (b) Show that the Jacobian for the dimensionless system is

$$J = \begin{bmatrix} 1 - 2x - \beta_1 y & -\beta_1 x \\ -\beta_2 y & \rho(1 - 2y) - \beta_2 x \end{bmatrix}.$$

- i. Determine the linear stability of equilibrium $e_1 = (1, 0)$ and sketch the phase portraits nearby. Are there any cases where the linear stability cannot tell us about the nonlinear stability of this equilibrium? Relate your analysis back to foxes and coyotes.
ii. Determine the linear stability of equilibrium $e_2 = (0, 1)$ and sketch the phase portraits nearby. Are there any cases where the linear stability cannot tell us about the nonlinear stability of this equilibrium? Relate your analysis back to foxes and coyotes.
iii. Determining the linear stability of equilibrium e_4 (the crossing point) is messy without a computer. Instead, use the nullclines around this point to determine the stability when $\beta_1 < 1$ and $\beta_2 < \rho$, and when $\beta_1 > 1$ and $\beta_2 > \rho$.
- (c) Conceptual questions about the model:
- i. What are some assumptions that are explicit or implicit in this model?
ii. Briefly describe each of the terms in the model. Which terms give the intraspecies competition? Which give the interspecies competition?

- iii. Which parameters are likely to be hard to measure/constrain? Which would be easy to measure/constrain?
- (d) Conceptual questions about results/analysis:
- i. In New York's Adirondack region, coyotes prey on red foxes and the NYS Department of Environmental Conservation reports that red foxes "avoid coyote territories completely or reside on the periphery of established coyote territories." Is there any inconsistency with this fact and the model we just analyzed? If no, explain why. If yes, can you adjust the model to fix it?
 - ii. What in the above analysis would you change if we replaced the red foxes with bobcats? The Adirondack Ecological Center site on bobcats will be helpful: <https://www.esf.edu/aec/adks/mammals/bobcat.htm>.
 - iii. What in the above analysis would you change if we replaced the coyotes with gray foxes? The Adirondack Ecological Center site on gray foxes will be helpful: https://www.esf.edu/aec/adks/mammals/gray_fox.htm.

Solution. Put

$$F(x, y) = x(1 - x) - \beta_1 xy, \quad G(x, y) = \rho y(1 - y) - \beta_2 xy.$$

- (a) Since $N_1 = A_1 x$, $N_2 = A_2 y$, and $t = T\tau$,

$$\frac{dN_1}{dt} = \frac{A_1}{T} \frac{dx}{d\tau}, \quad \frac{dN_2}{dt} = \frac{A_2}{T} \frac{dy}{d\tau}.$$

Substitution into the dimensional equations and division by A_1/T and A_2/T , respectively, yields

$$\begin{aligned} \frac{dx}{d\tau} &= Tr_1 x \left(1 - \frac{A_1}{K_1} x \right) - T\alpha_1 A_2 xy, \\ \frac{dy}{d\tau} &= Tr_2 y \left(1 - \frac{A_2}{K_2} y \right) - T\alpha_2 A_1 xy. \end{aligned}$$

Choose

$$A_1 = K_1, \quad A_2 = K_2, \quad T = \frac{1}{r_1}.$$

Then

$$\frac{dx}{d\tau} = x(1 - x) - \beta_1 xy, \quad \frac{dy}{d\tau} = \rho y(1 - y) - \beta_2 xy,$$

with the stated definitions of ρ , β_1 , and β_2 .

- i. Factoring gives

$$F = x(1 - x - \beta_1 y), \quad G = y(\rho(1 - y) - \beta_2 x).$$

Hence

$$\boxed{x = 0 \text{ or } y = \frac{1 - x}{\beta_1}}, \quad \boxed{y = 0 \text{ or } y = 1 - \frac{\beta_2}{\rho} x}.$$

- ii. The parameter-independent equilibria are

$$e_0 = (0, 0), \quad e_1 = (1, 0), \quad e_2 = (0, 1).$$

They represent extinction of both species, foxes alone at their carrying capacity $N_1 = K_1$, and coyotes alone at their carrying capacity $N_2 = K_2$, respectively.

iii. Solving the two oblique-nullcline equations gives

$$e_4 = (x_*, y_*), \quad x_* = \frac{\rho(1 - \beta_1)}{\rho - \beta_1\beta_2}, \quad y_* = \frac{\rho - \beta_2}{\rho - \beta_1\beta_2}.$$

This point is strictly in the first quadrant precisely when

$$\boxed{\beta_1 < 1 \text{ and } \beta_2 < \rho} \quad \text{or} \quad \boxed{\beta_1 > 1 \text{ and } \beta_2 > \rho}.$$

In the first regime, each species limits itself more strongly than it limits the other species, so stable coexistence is possible. In the second, interspecific effects are relatively stronger; the crossing still exists, but it separates the basins in a competitive exclusion regime. If $\beta_1 = 1$ and $\beta_2 = \rho$, the oblique nullclines coincide and there is a continuum of equilibria rather than exactly one crossing.

(b) Direct differentiation gives

$$J(x, y) = \begin{bmatrix} 1 - 2x - \beta_1 y & -\beta_1 x \\ -\beta_2 y & \rho(1 - 2y) - \beta_2 x \end{bmatrix}.$$

i. At $e_1 = (1, 0)$,

$$J(e_1) = \begin{bmatrix} -1 & -\beta_1 \\ 0 & \rho - \beta_2 \end{bmatrix}, \quad \lambda_1 = -1, \quad \lambda_2 = \rho - \beta_2.$$

Thus e_1 is a stable node if $\beta_2 > \rho$ and a saddle if $\beta_2 < \rho$. Biologically, a rare coyote population cannot invade the fox-only state when $\beta_2 > \rho$, but can invade when $\beta_2 < \rho$. When $\beta_2 = \rho$, one eigenvalue is zero and linearization is inconclusive. A center-manifold calculation gives $y' = \rho(\beta_1 - 1)y^2 + O(y^3)$ in the biologically relevant side: e_1 is attracting from the interior if $\beta_1 < 1$, repelling in the center direction if $\beta_1 > 1$, and belongs to a line of equilibria if also $\beta_1 = 1$.

ii. At $e_2 = (0, 1)$,

$$J(e_2) = \begin{bmatrix} 1 - \beta_1 & 0 \\ -\beta_2 & -\rho \end{bmatrix}, \quad \lambda_1 = 1 - \beta_1, \quad \lambda_2 = -\rho.$$

Hence e_2 is a stable node if $\beta_1 > 1$ and a saddle if $\beta_1 < 1$. A rare fox population cannot invade the coyote-only state when $\beta_1 > 1$, but can invade when $\beta_1 < 1$. For $\beta_1 = 1$, linearization is inconclusive. Near the center direction, $x' = (\beta_2/\rho - 1)x^2 + O(x^3)$, so the point attracts from the interior if $\beta_2 < \rho$, repels in that direction if $\beta_2 > \rho$, and lies on a line of equilibria when $\beta_2 = \rho$ as well.

iii. At the interior equilibrium, the nullcline identities reduce the Jacobian to

$$J(e_4) = \begin{bmatrix} -x_* & -\beta_1 x_* \\ -\beta_2 y_* & -\rho y_* \end{bmatrix}.$$

Therefore

$$\text{tr } J(e_4) = -(x_* + \rho y_*) < 0, \quad \det J(e_4) = x_* y_* (\rho - \beta_1 \beta_2).$$

If $\beta_1 < 1$ and $\beta_2 < \rho$, the determinant is positive, so e_4 is asymptotically stable. If $\beta_1 > 1$ and $\beta_2 > \rho$, the determinant is negative, so e_4 is a saddle; its stable

manifold separates initial conditions leading to fox exclusion from those leading to coyote exclusion.

- (c) i. The model assumes a closed, well-mixed, spatially homogeneous system; continuous deterministic populations; constant parameters; logistic growth for each species in isolation; instantaneous, pairwise, density-dependent competition; no age or sex structure; no migration, time delay, seasonality, disease, prey dynamics, or environmental noise; and fixed carrying capacities.

ii. In

$$r_i N_i \left(1 - \frac{N_i}{K_i}\right) = r_i N_i - \frac{r_i}{K_i} N_i^2,$$

$r_i N_i$ is density-independent growth and $-(r_i/K_i)N_i^2$ is intraspecific competition. The terms $-\alpha_1 N_1 N_2$ and $-\alpha_2 N_1 N_2$ are the effects of interspecific encounters on foxes and coyotes, respectively.

- iii. Abundance data and approximate low-density growth rates may be comparatively accessible, while carrying capacities remain habitat- and season-dependent. The interaction coefficients α_1 and α_2 are usually hardest to identify because they require separating interspecific effects from habitat, resources, predation, and intraspecific density dependence. The dimensionless parameters inherit this uncertainty.
- (d) i. There is a modeling mismatch if the equations are interpreted literally as a spatial description: a well-mixed ODE has no territories and cannot represent avoidance. It can still describe average abundance effects after spatial behavior has been folded into the α_i . Also, coyote predation is not the same as symmetric competition: a predator-prey formulation would typically reduce fox growth but increase coyote growth, for example replacing the second interaction term by $+\gamma N_1 N_2$ together with an appropriate coyote mortality/resource model. Territories can be represented using spatial patches, diffusion/advection, or density-dependent encounter rates.
- ii. Replacing red foxes by bobcats changes the parameter values and weakens the justification for a simple well-mixed competition model. Bobcats are solitary, use very large home ranges, are almost entirely carnivorous, and show strong habitat and seasonal effects; coyote interactions may also be important. A better model could include spatial habitat classes, seasonal carrying capacities, shared-prey dynamics, and asymmetric competition. The algebraic stability analysis is unchanged only if one retains the same two-species competition equations.
- iii. Replacing coyotes by gray foxes also preserves the algebra only under the same modeling assumptions. Gray foxes use brushy and deciduous woodland more than the open habitats preferred by red foxes, and their opportunistic diet overlaps only partly with that of red foxes. Habitat partitioning should therefore reduce effective encounter rates or require separate patches. Because coyotes may prey on young gray foxes, any retained coyote effect would be asymmetric rather than pure competition. Thus one should re-estimate $r_2, K_2, \alpha_1, \alpha_2$ and consider explicit habitat and predation terms.

□